

MOHAMED AGHNAOU



Software Engineer seeking a full-time position.

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Software Development Engineer, with a strong background in designing and building end-to-end software solutions. Skilled in backend and frontend development, database design, and application integration. Experienced in developing scalable, maintainable applications and working across the full software lifecycle.

Education

- 📅 2022–2025

Software Development Engineering
[National Institute of Posts and Telecommunications \(INPT\)](#)
- 📅 2020–2022

Preparatory Classes for Engineering Schools (MP)
[Lycée Réda Slaoui - Agadir](#)

Professional Experience

ORACLE

📅 01/2025–09/2025 📍 Casablanca, Morocco

[Software Engineer – Research Assistant](#)

- Designed and implemented a backend feature enabling users to unlink Intercompany Cross Charge Journals (ICCs) directly from the NetSuite interface.

- Ensured financial data integrity by enforcing accounting period constraints, permission rules, and guaranteeing atomicity of database operations.

- Established a comprehensive testing strategy including unit, component, and E2E tests, ensuring reliability of the feature in a production environment.

Technologies : Java, NetSuite ERP, Git, GitLab, Jira, TeamCity, Oracle Database, SuiteQL, JUnit, Mockito

Royal Air Maroc

📅 06/2024–08/2024 📍 Casablanca, Morocco

[Full-Stack Engineer](#)

- Developed a web application promoting sustainable tourism and natural travel destinations.

- Implemented interactive eco-travel exploration and booking features based on dynamic itineraries.

- Designed secure user authentication with real-time reviews.

- Integrated secure online payment processing for reservations.

Technologies : NodeJs, Express, MongoDB, Mongoose, Pug, Git, Github

Projects

AI-Powered Enterprise Knowledge Management Assistant

08/2025–10/2025

Development of an AI-based application enabling users to query enterprise documentation in natural language:

- Provide contextual answers from unstructured internal documents

- Implement a **RAG** mechanism to enrich queries with relevant information

- Develop an AI agent using the **MCP protocol** to dynamically orchestrate multiple tools (enterprise data, file system, external services)

Technologies: Java, Spring Boot, Spring AI, OpenAI API, MCP, AI Agent, RAG, REST API, Python, Node.js, Angular

BiblioManager Application

03/2024–05/2024

Developed an application simplifying book and reservation management:

- Implemented an intuitive system enabling users to add, search, and track available books.

- Developed reservation management and user review features.

Technologies : ReactJs, Bootstrap, Spring Boot, Spring Security, Spring Data JPA, Spring JWT, Okta

ReserveINPT

06/2023–08/2023

Developed a mobile reservation application for managing institutional facilities:

- Enabled users to create, modify, and cancel reservations.

- Provided administrative tools for managing student accounts and facilities.

Technologies : Java, AndroidStudio

Skills

- Programming :** Java, JavaScript, TypeScript, Python, Kotlin
- Backend Development :** JEE, Spring, Spring Boot, Spring Data, Spring Cloud, Spring Security, Keycloak, NodeJs, Expresss
- Frontend Development :** HTML, Bootstrap, Tailwind CSS, ReactJs, Angular, Next.js
- Generative AI & LLMs :** LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Multimodal AI, OpenAI API, LLaMA
- AI Agents & Automation :** Agentic AI, Model Context Protocol (MCP), Spring AI, Workflow Orchestration, Tool Calling
- DevOps :** Docker, Kubernetes, Kafka, GitLab CI/CD, Jenkins
- Databases :** MySQL, SQL Server, MongoDB, Oracle Database, PostgreSQL
- Software Architectures :** Microservices, Event Driven Architecture,Event Sourcing, CQRS, Saga Pattern OAuth2.0, REST APIs, GraphQL, DDD

Languages

Arabic : Native **English :** Professional **French :** Fluent (DELF B2)

Soft Skills

Communication - Teamwork - Problem Solving - Adaptability - Self-learning